Universitè Grenoble Alpes - INP

Master of Science in Industrial and Applied Mathematics

Computational Biology

Design and Implement simple individual-based simulations of evolution

Alibrandi Orazio, Guarnaschelli Eleonora

Academic year: 2025-2026

Contents

1 Introduction **3**

2 Model Specification **4**
2.1 Population structure and assumptions 4
2.2 Phenotypic representation and fitness landscape 4
2.3 Demographic dynamics: Non-Neutral Moran process 6
2.4 Mutation process 6

3 Experiments **7**
3.1 Experimental design choices 7
3.2 Selection for or against mutator alleles 8
3.3 Effect of mutator strength 11
3.3.1 Isolated Effect of α 11
3.3.2 Mutator fixation probability as a function of mutation rate 14
3.4 Punctuated Equilibrium 15
3.5 Effect of phenotypic complexity with fixed pleiotropy 16
3.6 Dynamics of a Moving Optimum 17
3.6.1 Geometry of the Movement 17
3.6.2 Stepwise Dynamics 17
3.6.3 Simulation Parameters 17
3.6.4 Results 18

4 Conclusion **20**

References **21**

1 Introduction

Asexual microbial populations provide a powerful framework for studying evolutionary dynamics, as their short generation times, large population sizes, and lack of recombination allow evolutionary processes to be observed over experimentally accessible timescales. In such systems, mutations remain permanently linked to the genetic background in which they arise, giving rise to stochastic evolutionary trajectories shaped by selective sweeps and random genetic drift.

A central question in this context concerns the evolution of mutation rates. Although most mutations are deleterious, alleles that increase the mutation rate (mutator alleles) are frequently observed in microbial populations. From a purely selective perspective, mutators are expected to be disfavored due to their increased deleterious load. Nevertheless, mutator alleles may spread indirectly by generating beneficial mutations at a higher rate than wild-type individuals. When a beneficial mutation arises and establishes in a mutator background, the mutator allele can increase in frequency together with it through genetic hitchhiking, despite having no intrinsic selective advantage.

This mechanism was experimentally investigated by Chao and Cox [1], who studied competitions between *Escherichia coli* strains differing only in their mutation rate. Their experiments showed that the evolutionary fate of a mutator allele depends strongly on its initial frequency: above a critical threshold, mutators typically reach fixation after a characteristic delay, whereas below this threshold they are usually eliminated. This delay reflects the stochastic waiting time for the appearance and successful establishment of a beneficial mutation of sufficient effect.

Starting from this experimental setting, we ask whether such dynamics can be reproduced and explored within a minimal theoretical framework, and whether this framework can be extended to investigate additional evolutionary phenomena. In particular, we focus on the effects of increased mutation rates, the emergence of punctuated equilibrium dynamics, environmental changes, and increasing phenotypic dimensionality.

To this end, we develop an individual-based model of an asexual population evolving under a non-neutral Moran process. Individuals are represented as points in an abstract phenotypic space and mapped to fitness through Fisher's Geometrical Model (FGM), in which selection favors phenotypes closer to an environmental optimum. This formulation naturally generates both beneficial and deleterious mutations. In addition, individuals carry a binary mutator allele that controls their mutation rate, allowing us to study the coupled dynamics of adaptation and mutation-rate evolution.

Overall, this project employs a minimal yet flexible modeling framework to connect classical experimental results with fundamental concepts in evolutionary theory, highlighting the interplay between stochasticity, mutation, and selection in finite asexual populations.

2 Model Specification

In this section, we describe in more technical detail the evolutionary model developed in this project.

2.1 Population structure and assumptions

We consider a constant-size population of N individuals evolving in discrete time under the following assumptions:

1. Reproduction is strictly asexual and clonal; recombination is absent.
2. Individuals belong to one of two genetic backgrounds:
   - wild type ($m = 0$),
   - mutator ($m = 1$), characterized by an increased mutation rate.
3. The mutator allele does not directly affect fitness or the fitness landscape; its only effect is to scale the mutation rate.
4. Mutations can be beneficial, neutral, or deleterious depending on their phenotypic effect.
5. Population size is kept constant through a non-neutral Moran process.

2.2 Phenotypic representation and fitness landscape

Individuals are described by an abstract phenotypic representation rather than by explicit genotypes. Each individual i is characterized by a phenotypic vector

$$\mathbf{z}_i \in \mathbb{R}^d,$$

where d denotes the dimensionality of the phenotypic space, interpreted as the number of genetically uncorrelated traits exposed to natural selection. This definition follows the notion of phenotypic complexity introduced in [2].

The environment is defined through an optimal phenotype $\mathbf{z}^*$, which represents the combination of trait values that maximizes fitness under given environmental conditions. The concept of a fitness landscape, originally introduced by Sewall Wright, provides a useful geometrical interpretation: each phenotype corresponds to a point in trait space, while fitness is represented as the height of the landscape at that point. Evolution can thus be interpreted as the movement of a population across this landscape, driven by mutation and selection. Fitness landscapes can be single-peaked or rugged; in this work, we focus on a single-peaked landscape. Fitness is assigned according to Fisher's Geometrical Model, in which the phenotype–fitness map depends solely on the distance between an individual's phenotype and the environmental optimum. In particular, fitness depends on the Euclidean distance

$$r = \|\mathbf{z} - \mathbf{z}^*\|$$

between the phenotype and the optimum, and is modeled using a Gaussian stabilizing selection function,

$$W(r) = \exp\left(-\frac{r^2}{2\sigma^2}\right),$$

where σ controls the width of the fitness peak and thus the strength of stabilizing selection.

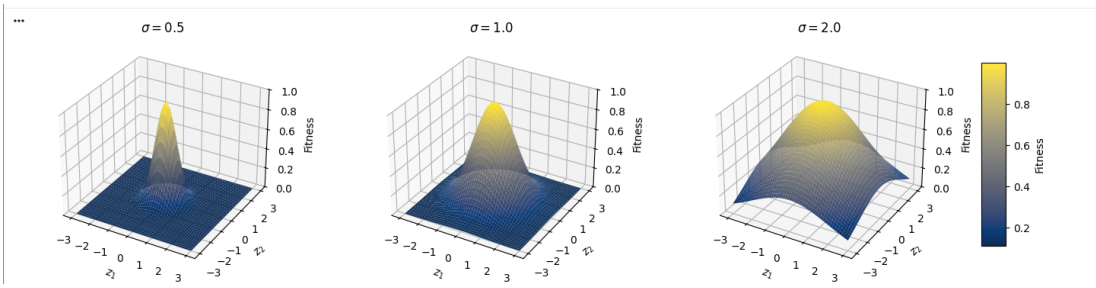


Figure 2.1: Fisher's geometric model for different values of σ. Smaller values of σ correspond to narrower fitness peaks and stronger stabilizing selection, while larger values produce broader peaks.

Individual fitness determines the probability that an individual survives and reproduces in the next generation. Evolutionary dynamics are therefore naturally described by the trajectories of phenotypic points in a d-dimensional space over time.

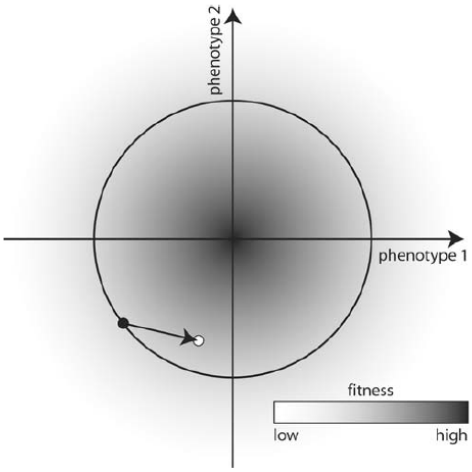


Figure 2.2: Fisher's geometric model in two-dimensional phenotypic space. Fitness varies along two phenotypic axes, with the maximum fitness located at the optimum. Fitness form circles (in 2D) centered on the optimum. Mutations correspond to random displacements in phenotype space. [2].

Within this framework, mutations generate offspring whose phenotypes differ from those of their parents by random displacement vectors. A mutation is considered beneficial if it moves the phenotype closer to the environmental optimum $\mathbf{z}^*$, increasing fitness, and deleterious otherwise. Importantly, the same mutational displacement may have different fitness effects depending on the initial phenotypic position, reflecting the context-dependent nature of mutational effects.

Mutational effects are assumed to be isotropic and unbiased, implying that mutations have no preferred direction in phenotype space and that, conditional on being affected, all phenotypic dimensions are subject to identical mutational effects. In this project, mutational pleiotropy is modeled by allowing each mutation to affect only a subset of k phenotypic traits chosen at random from the total d traits. Universal pleiotropy is recovered as the special case $k = d$, while partial pleiotropy is assumed when exploring increasing phenotypic complexity, reflecting the biological plausibility that mutations are unlikely to affect all traits simultaneously when d is large [2]. Under these assumptions, mutations of small phenotypic effect are more likely to be beneficial than mutations of large effect, providing a natural explanation for gradual adaptive evolution. As phenotypic complexity increases, the fraction of beneficial mutations rapidly decreases, since in high-dimensional spaces the proportion of random

displacements pointing toward the optimum becomes very small.

Despite its apparent simplicity, Fisher's Geometrical Model generates rich evolutionary behavior. Epistatic interactions between mutations emerge naturally from the geometry of the fitness landscape, as the fitness effect of a mutation depends on the current phenotypic position. While the classical formulation of FGM assumes a fixed environmental optimum, the framework can be extended to incorporate temporal environmental change by allowing the optimum $\mathbf{z}^*$ to vary over time, a feature that is particularly relevant for studying adaptation in changing environments. More in detail, as in [3], we model gradual environmental change by assuming a linearly moving phenotypic optimum. The optimal phenotype evolves according to

$$\mathbf{z}^*(t) = \mathbf{v}\, t,$$

so that environmental change is characterized by a constant velocity $\mathbf{v}$ in phenotype space, which determines both the speed and direction of adaptation.

2.3 Demographic dynamics: Non-Neutral Moran process

Evolution proceeds through a non-neutral Moran process:

1. One individual is chosen to reproduce with probability proportional to its fitness.
2. One individual is chosen uniformly at random to be removed.
3. The "child" inherits the phenotype and mutator status of the parent, subject to mutation.

This process keeps the population size constant while allowing selection to act continuously.

2.4 Mutation process

Mutations occur during reproduction with probability:

$$\mu = \begin{cases} \mu_{\mathrm{WT}} & \text{if } m = 0, \\ \alpha\mu_{\mathrm{WT}} & \text{if } m = 1, \end{cases}$$

where $\alpha \gg 1$ represents the strength of the mutator allele.

When a mutation occurs, a random subset of k phenotypic dimensions is affected. For the affected traits, mutational effects are drawn independently from a normal distribution with mean zero and variance η^2, while all other traits remain unchanged. The phenotype is thus updated as

$$\mathbf{z}' = \mathbf{z} + \boldsymbol{\delta}.$$

Small values of η produce small-effect mutations that are more likely to be beneficial or nearly neutral, while large η generate predominantly deleterious mutations.

Parameter	Meaning
N	Population size
d	Phenotypic complexity
$\mathbf{z}$	Phenotypic vector
k	Mutational pleiotropy
$f_M(0)$	Initial mutator frequency
$\mathbf{z}^*$	Optimal phenotype
σ	Width of the fitness peak
μ_{WT}	Baseline mutation rate
α	Mutator mutation-rate multiplier
η^2	Variance of mutational effects
w_0	Initial fitness

Table 1: Summary of model parameters.

3 Experiments

In this section we describe the main experimental design choices adopted to explore mutator dynamics under Fisher's Geometrical Model and the obtained results for each phenomena investigated.

3.1 Experimental design choices

Time scale of simulations Evolutionary dynamics are simulated using an individual-based Moran birth-death process. Each Moran time step corresponds to a single birth-death event. To facilitate comparison with previous experimental and theoretical studies reporting time in generations [1], simulation time is converted into generations as

$$G = \frac{t}{N}, \tag{3.1}$$

so that one generation corresponds approximately to N Moran steps.

Each Moran step is implemented through a single call to the `moran_step` routine, which performs reproduction, mutation, and replacement, and updates individual phenotypes, mutator status, and (when applicable) the position of the environmental optimum.

Environmental scenarios Two environmental regimes are considered. In the static environment, the phenotypic optimum remains fixed throughout the simulation. In the dynamic environment, the optimum moves linearly in phenotypic space, following the moving-optimum framework of [3].

To reflect biologically realistic environmental change, the position of the optimum is updated every 5000 Moran steps rather than continuously at every step, ensuring that environmental shifts occur on a slower timescale than individual birth-death events.

Beneficial mutations and stopping criteria A mutation is classified as beneficial if the fitness of the mutant exceeds the mean population fitness prior to the Moran update. This definition captures the immediate selective advantage of a mutation independently of demographic stochasticity.

Simulations can be terminated early when the mutator allele reaches fixation or extinction. When early stopping is enabled, fixation of the mutator reflects successful hitchhiking with beneficial mutations arising on the mutator background, while extinction corresponds to competitive exclusion by the wild type. This stopping criterion alows us to focus specifically on mutator–wild-type competition rather than on long-term convergence to the phenotypic optimum.

Initial conditions and dimensionality control When comparing different values of phenotypic dimensionality d, simulations are initialized at the same adaptive distance from the fitness optimum. This is achieved by choosing the position of the optimum such that the initial population has identical mean fitness w_0 across all dimensionalities, with the distance to the optimum distributed equally along each phenotypic axis.

This choice ensures that observed differences across dimensionalities are not confounded by differences in initial bad adaptation. When w_0 is not specified, the optimum is set to $z^* = (1, \ldots, 1)$.

Recorded observables For each simulation, we record:

- the mutator allele frequency $f_M(t)$,
- the mean population fitness over time,
- detailed mutation event histories, including selective effects relative to both the parent and the population.

Unless otherwise stated, all parameters are kept fixed to the baseline values reported in Table 2.

Parameter	Meaning	Value
N	Population size	1000
d	Phenotypic complexity	2
k	Mutational pleiotropy	2
$f_M(0)$	Initial mutator frequency	40/1000
σ	Width of the fitness peak	1
μ_{WT}	Baseline mutation rate	2×10^{-4}
α	Mutator mutation-rate multiplier	200
η^2	Variance of mutational effects	0.1
w_0	Initial fitness	0.8

Table 2: Summary of model parameters.

In most simulations, phenotypic dimensionality is fixed to $d = 2$. This choice is motivated by experimental estimates suggesting that the effective phenotypic complexity of *E. coli* is low, typically between two and three [2]. Higher-dimensional simulations are performed only when explicitly investigating the effect of increasing phenotypic complexity.

3.2 Selection for or against mutator alleles

Following [1], we study the competition between a mutator strain and a wild-type strain in an asexual population. Chao and Cox showed that the selective advantage of a mutator allele is strongly frequency dependent: when introduced at sufficiently low initial frequency, the mutator population is typically unable to displace the wild-type population. Moreover, their experiments revealed the existence of a characteristic lag phase preceding the rapid increase of the mutator population. This advantage does not arise from an intrinsic growth-rate benefit, but from the increased production of beneficial mutations in the mutator background.

Here, we investigate whether the same qualitative behavior emerges in our individual-based simulations.

Let $f_M(t)$ denote the mutator allele frequency at evolutionary time t, so that the wild-type frequency is $1 - f_M(t)$. As in [1], we characterize the dynamics using the log-ratio

$$R(t) \;=\; \log_{10}\left(\frac{f_M(t)}{1 - f_M(t)}\right), \tag{3.2}$$

In practice, $R(t)$ is computed from the simulated mutator frequency time series, with a small numerical regularization to avoid divergences when $f_M(t)$ approaches 0 or 1.This regularization reflects the finite resolution imposed by population size and does not affect the qualitative behavior of the trajectories.

For each value of the initial mutator frequency $f_{0,M}$, multiple independent simulations are performed with different random seeds. Fixation probabilities are estimated empirically as the fraction of simulations in which the mutator allele reaches fixation, and uncertainty is quantified using Wilson confidence intervals. To visualize typical evolutionary dynamics, we do not average trajectories across replicates. Instead, for each value of $f_{0,M}$ we select a representative realization corresponding to the most probable outcome (fixation or extinction). The representative trajectory is chosen as the simulation whose absorption time is closest to the median absorption time among realizations leading to the same outcome. This procedure avoids bias toward atypically fast or slow trajectories and provides a faithful illustration of the typical competitive dynamics.

Empirically, $R(t)$ often displays two approximately linear regimes separated by a change in slope, corresponding to an initial lag phase followed by a rapid rise or decline. In said in [1], these regimes can be interpreted as reflecting the waiting time for the first successful beneficial mutation and its subsequent fixation.

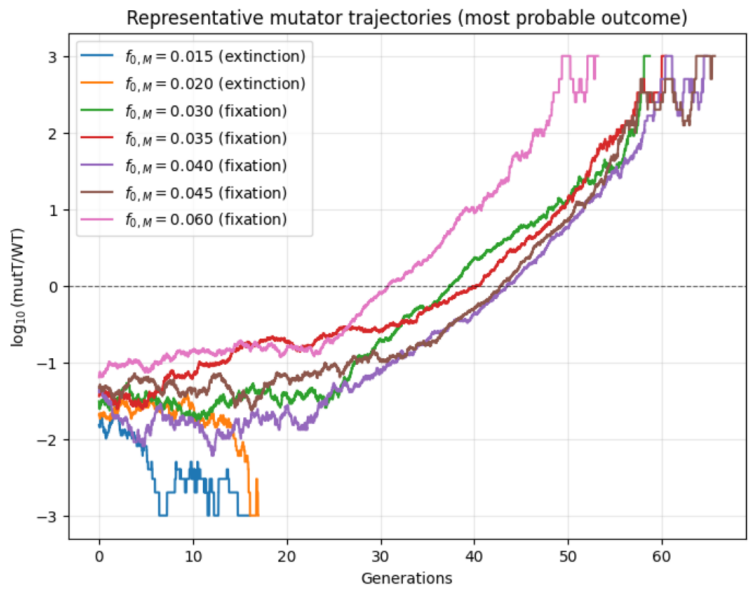


Figure 3.1: Representative trajectories of the log-ratio $R(t)$ for different initial mutator frequencies $f_{0,M}$.

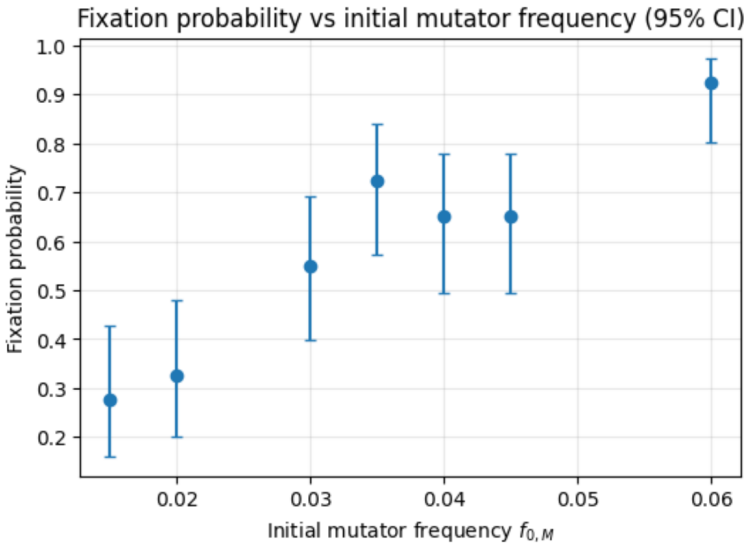


Figure 3.2: Estimated fixation probability of the mutator allele as a function of its initial frequency $f_{0,M}$. Error bars represent 95% Wilson confidence intervals.

Figures 3.1 and 3.2 show that mutator fixation is strongly frequency dependent. A transition region is observed around $f_{0,M} \approx 0.03$–0.04, where fixation and extinction occur with comparable probability. The corresponding trajectories reveal a clear lag phase of approximately 20 generations, followed by a rapid increase in $R(t)$ when fixation occurs.

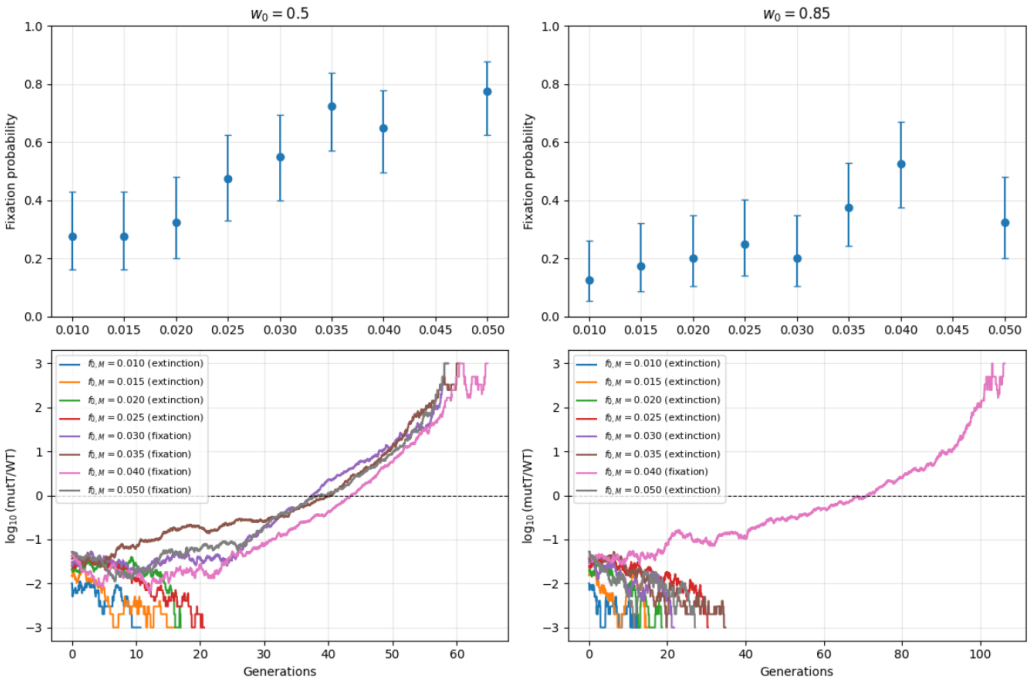


Figure 3.3: Effect of the initial mean fitness w_0 on mutator dynamics. Upper panels: fixation probability as a function of $f_{0,M}$. Lower panels: representative trajectories of $R(t)$ corresponding to the most probable outcome.

Figure 3.3 summarizes the outcome of this competition for two levels of initial adaptation, quantified by the initial mean fitness w_0. When the population starts farther from the fitness optimum ($w_0 = 0.5$),

the fixation probability increases monotonically with $f_{0,M}$. In this regime, beneficial mutations are several, allowing mutators to generate adaptive variants early enough to overcome stochastic loss. In contrast, when the population is initially well adapted ($w_0 = 0.85$), the fixation probability of the mutator allele is markedly reduced across the same range of $f_{0,M}$. This reflects the reduced availability of beneficial mutations near the fitness peak, which weakens the indirect selective advantage of the mutator allele.

Overall, these results reproduce the central qualitative findings of [1]: selection on mutation rate is frequency dependent and mediated by the supply of beneficial mutations that,in our simmulations, arise stochastically during evolution.

3.3 Effect of mutator strength

While several experimental and theoretical studies have shown that increasing the genomic mutation rate can accelerate adaptation by increasing the supply of beneficial mutations, it remains unclear whether this advantage persists at arbitrarily high mutation rates. A variety of models suggest that extremely high mutation rates may hinder adaptation by generating an excessive load of deleterious mutations. Early theoretical work predicted the existence of a critical "error threshold", beyond which population fitness declines as mutation rate increases, although the exact nature of this threshold depends on model assumptions [4].

3.3.1 Isolated Effect of α

Here, we investigate whether a similar trade-off emerges in an individual-based model combining a non-neutral Moran process with Fisher's Geometrical Model. In particular, we ask whether increasing the mutator strength α leads to diminishing returns or reduced adaptive success, even in the absence of an explicit fitness cost associated with the mutator allele. To isolate the effect of the mutation rate from competitive dynamics between genetic backgrounds, all simulations are initialized with the mutator allele fixed ($f_{0,M} = 1$), so that the entire population experiences the same mutation regime.

Adaptive dynamics are first characterized through the mean population fitness $\bar{w}(t)$. In addition, to quantify how efficiently the population exploits the available adaptive potential, we introduce the normalized relative fitness gain,

$$G(t) = \frac{\bar{w}(t) - w_0}{1 - w_0},$$

where w_0 denotes the initial mean fitness and 1 corresponds to the maximum accessible fitness at the phenotypic optimum. This quantity measures the fraction of the total possible fitness improvement that has been realized and therefore captures adaptive efficiency rather than absolute fitness alone.

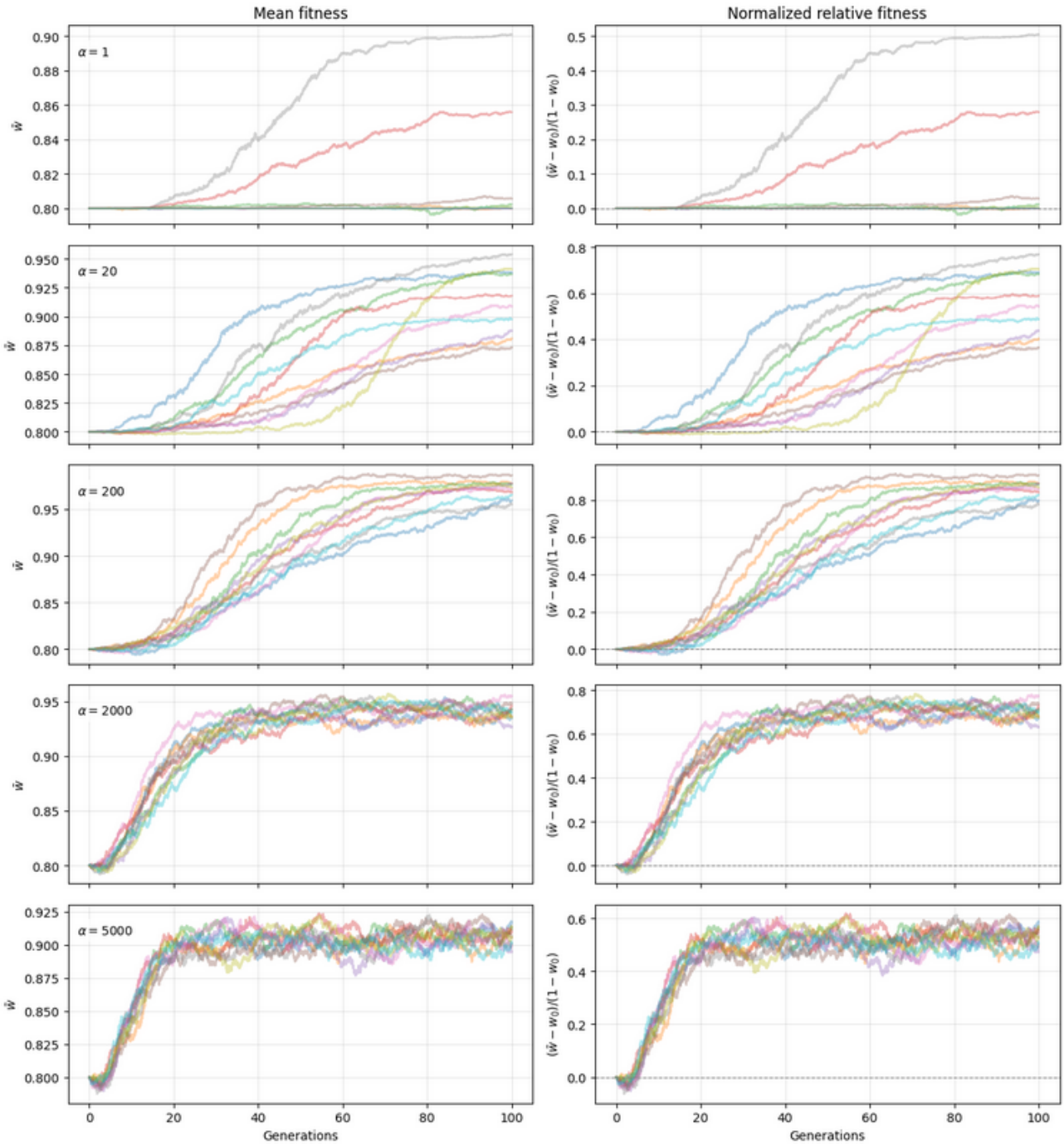
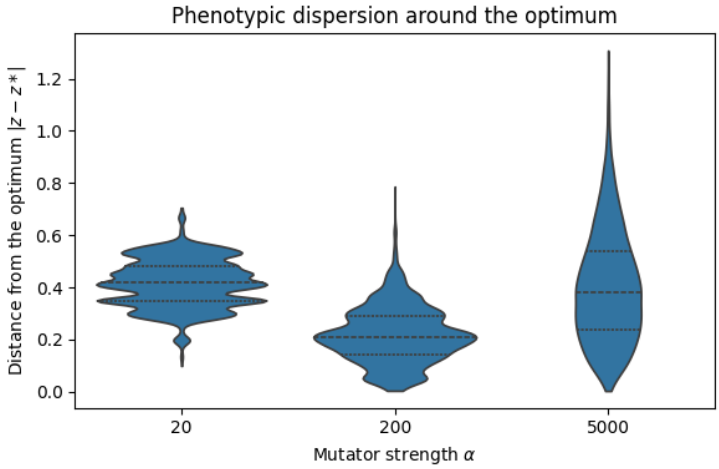


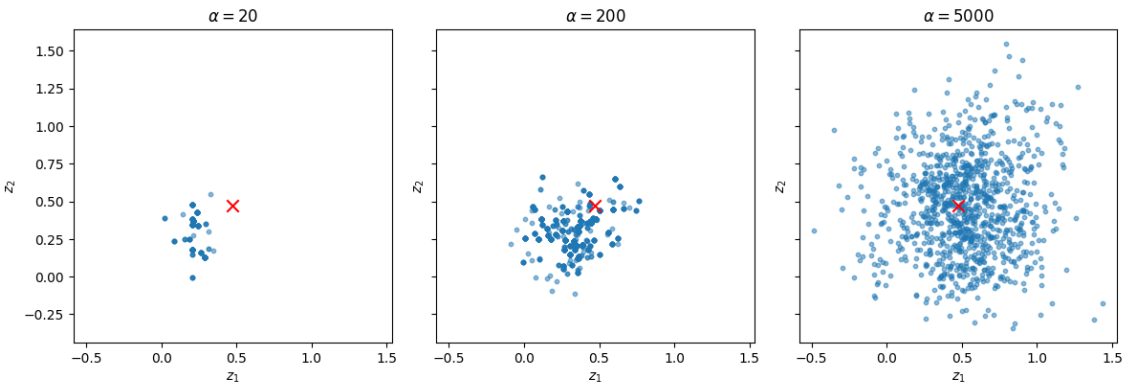
Figure 3.4: Mean fitness vs Gain normalized fitness for several values of α.

Figure 3.4 shows that increasing α strongly accelerates the early phase of adaptation: for large mutation rates the population rapidly approaches high fitness values and stabilizes within a few tens of generations, whereas for intermediate values of α adaptation proceeds more slowly and for smaller ones is not reached at all during the chosen time period. However, the normalized relative fitness gain reveals a qualitatively different behavior. While $\alpha = 200$ leads to a steady increase in $G(t)$, eventually exceeding 0.8, larger mutation rates result in progressively lower asymptotic gains. In particular, for $\alpha = 2000$ the normalized gain saturates at slightly lower values, and for $\alpha = 5000$ it stabilizes around $G \simeq 0.6$. This indicates that, despite achieving high mean fitness, populations evolving at very high

mutation rates fail to exploit a substantial fraction of the available adaptive potential.



(a) Distribution of phenotypic distances from the optimum for different values of α.



(b) Scatter plots of individual phenotypes in two-dimensional trait space at the final simulation time.

Figure 3.5: Phenotypic structure of populations evolving at different mutation rates. Intermediate mutation rates lead to a tight concentration around the fitness optimum, whereas very high mutation rates produce broad phenotypic dispersion.

The origin of this reduced adaptive efficiency becomes clear when examining the phenotypic structure of the population at the end of the simulations. Figure 3.5 shows that intermediate mutation rates produce a tight concentration of individuals around the fitness optimum, corresponding to a narrow distribution of phenotypic distances. In contrast, very high mutation rates generate a broad phenotypic cloud, with individuals widely dispersed in trait space and a long-tailed distribution of distances from the optimum. Although selection maintains a high average fitness, the population remains in a dynamic mutation–selection balance in which beneficial mutations are continuously counteracted by a large influx of deleterious ones.

Taken together, these results highlight a clear trade-off between the speed and the efficiency of adaptation. Increasing the mutation rate accelerates the approach to the fitness peak, but excessively high mutation rates prevent stable concentration near the optimum and reduce the fraction of adaptive potential that is effectively realized. In this framework, high-α populations are best described as persistently perturbed rather than stably adapted, providing an error-threshold like interpretation of mutation-rate effects even in a smooth, single-peaked fitness landscape.

3.3.2 Mutator fixation probability as a function of mutation rate

After characterizing how the mutation rate affects the adaptive dynamics of a fully mutator population, we next investigate how mutator strength influences the outcome of direct competition between mutator and wild-type lineages. In this set of simulations, the mutator allele is introduced at a fixed low initial frequency ($f_{0,M} = 0.035$), and its probability of fixation is measured as a function of the mutation-rate multiplier α.

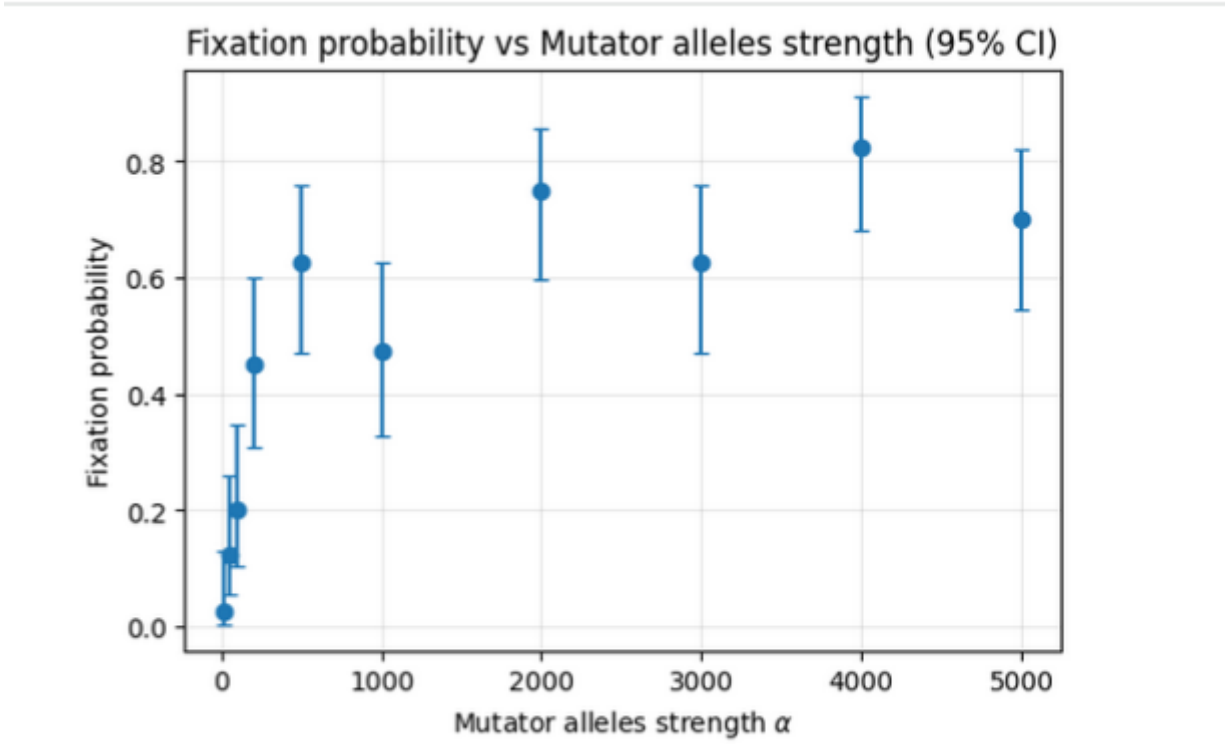


Figure 3.6: Mutator fixation probability as a function of mutator strength α. Error bars denote 95% Wilson confidence intervals.

Figure 3.6 shows the estimated fixation probability of the mutator allele across a wide range of α values. Fixation probabilities are obtained from repeated independent simulations and reported together with 95% Wilson confidence intervals. For small values of α, fixation is relatively unlikely, reflecting the limited supply of beneficial mutations generated on the mutator background. As α increases, fixation probability rises sharply, indicating that an elevated mutation rate enhances the likelihood that a mutator lineage generates a beneficial mutation early enough to escape stochastic extinction.

An interesting fact is that fixation probabilities for intermediate values of α (e.g. $\alpha = 200$ and $\alpha = 1000$) are statistically indistinguishable within confidence intervals, suggesting that mutator success saturates once the supply of beneficial mutations becomes sufficiently large. For larger values of α, fixation probability does not increase further and instead fluctuates in the range 0.6–0.8, indicating diminishing returns rather than a systematic decline.

In our implementation, $\alpha = 5000$ corresponds to a per-replication mutation probability $\mu_M = \alpha\mu_{\mathrm{WT}} \simeq 1$, implying that every mutator reproduction event results in a mutation. Consequently, exploring larger values of α would not increase the effective mutation rate further and is therefore not informative within this modeling framework.

To formally assess the dependence of fixation probability on mutator strength, we fitted logistic regression models using $\log_{10}(\alpha)$ as predictor and compared a linear model to a quadratic extension. A likelihood ratio test shows that the quadratic model provides a significantly better fit than the linear one ($p = 0.025$), providing statistical evidence for significant curvature in the relationship between fixation probability and mutation rate.

Taken together, these findings demonstrate that while increasing the mutation rate initially promotes mutator invasion by enhancing the supply of beneficial mutations, further increases lead to diminishing returns rather than continued gains in fixation probability. This result is consistent with a trade-off between beneficial mutation supply and the destabilizing effects of high mutational turnover, highlighting that mutator success is constrained not only by mutation availability but also by the stability of adaptive lineages in the presence of elevated mutational noise.

3.4 Punctuated Equilibrium

According to [5], two major schools of thought exist in evolutionary biology. Gradualism assumes that evolutionary change proceeds continuously over time, whereas the theory of punctuated equilibrium (PE) proposes that evolution occurs in bursts, with long periods of relative stasis interrupted by short episodes of rapid change.

Punctuated growth may arise from the action of natural selection on rare beneficial mutations. During periods of stasis, populations can be viewed as waiting for such mutations to occur, implying that the rate of evolution is limited by their availability rather than by selection alone.

In this project, we investigate the validity of this hypothesis within our simplified evolutionary framework based on Fisher's Geometrical Model. We consider a large, constant-size, asexual population evolving on a static fitness landscape, where spontaneous mutations are the sole source of genetic variation. The population is initialized far from the phenotypic optimum, and evolution proceeds through the accumulation of beneficial mutations that increase mean population fitness. Fitness is measured as the average reproductive success of individuals and is tracked over time in units of generations.

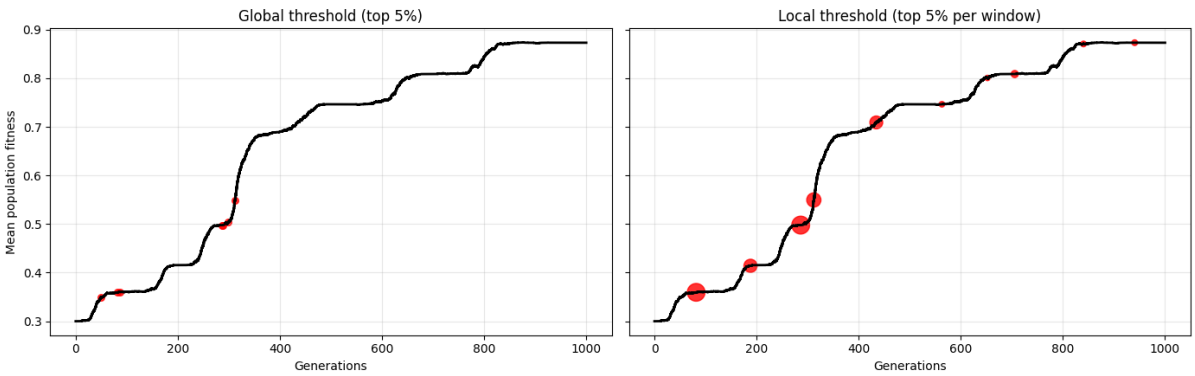


Figure 3.7: Punctuated equilibrium dynamics in the Fisher Geometrical Model. Mean population fitness exhibits long periods of stasis interrupted by rapid adaptive jumps driven by rare beneficial mutations.

Figure 3.7 illustrates the resulting adaptive dynamics. Mean population fitness displays extended plateaus followed by sudden increases, qualitatively reproducing the punctuated pattern expected under PE. Red markers indicate beneficial mutations selected according to two different criteria. In the left panel, mutations are identified using a global threshold, corresponding to the top 5% of beneficial effects across the entire evolutionary trajectory. This criterion highlights only early, large-effect mutations

and gives the misleading impression that adaptive evolution effectively slows down or ceases at later times.

However, fitness increases are still observed at late generations. To account for this, we introduce a local selection criterion, shown in the right panel, based on selecting the top 5% of beneficial mutations within sliding temporal windows of 100 generations. This representation reveals that, at every stage of adaptation, rare beneficial mutations relative to the current genetic background are responsible for the observed fitness jumps. Although the absolute magnitude of fitness effects decreases as the population approaches the optimum, these locally extreme events continue to drive evolutionary change. The decreasing size of the red markers reflects the progressive reduction in available beneficial effect sizes, in agreement with theoretical expectations from Fisher's Geometrical Model.

Overall, these results support the interpretation that punctuated equilibrium can emerge naturally from simple evolutionary dynamics in a constant environment, without invoking environmental change or additional genetic mechanisms. Long periods of stasis arise from the scarcity of beneficial mutations, while rapid adaptive bursts correspond to the fixation of rare, stochastic events, closely paralleling experimental observations in long-term evolution studies.

3.5 Effect of phenotypic complexity with fixed pleiotropy

We next investigate the effect of phenotypic complexity on adaptive dynamics by varying the dimensionality d of the phenotypic space, maintaining the pleiotropy equal to 3. Within the Fisher geometric framework, increasing d corresponds to increasing the number of traits under stabilizing selection, which is expected to reduce both the availability and the average effect of beneficial mutations.

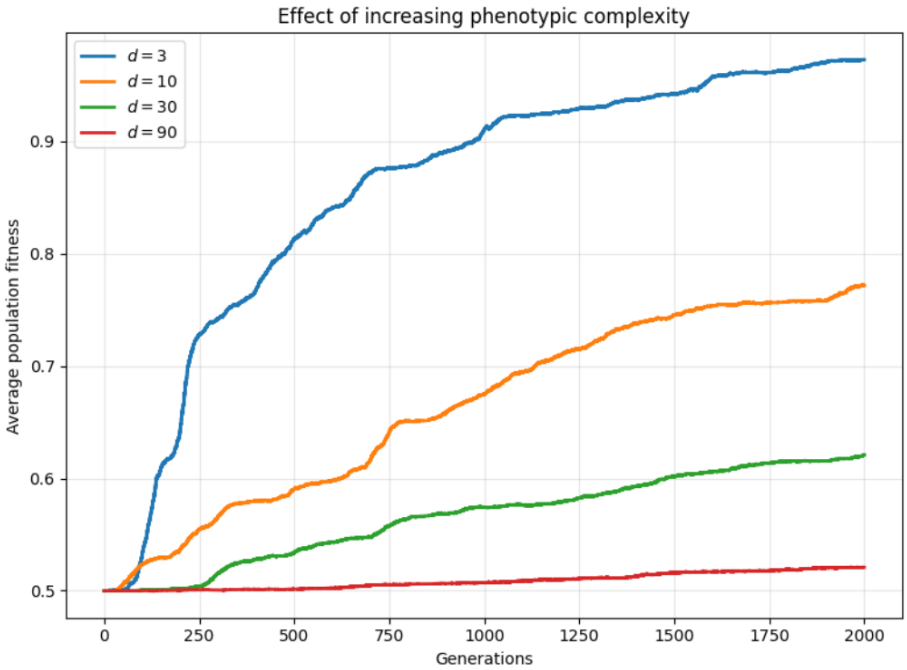


Figure 3.8: The effect of increasing phenotypic complexity d on the dynamics of fitness increase.

Figure 3.8 shows the time evolution of the mean population fitness for different values of d, starting from the same initial fitness $w_0 = 0.5$. Populations with lower phenotypic dimensionality exhibit a faster and more pronounced increase in fitness, reaching substantially higher fitness values within the simulated time window. In contrast, as d increases, adaptation becomes progressively slower and the fitness trajectories flatten, indicating a reduced adaptive potential.

This behavior is consistent with teoretical expectations and previous results [6]. As phenotypic complexity increases, random mutational steps are more likely to move the phenotype away from the optimum in at least one dimension, thereby decreasing the fraction of beneficial mutations. Consequently, adaptive substitutions become rarer and the rate of fitness increase declines.

3.6 Dynamics of a Moving Optimum

In this section, we present the results of simulations performed with a time-varying phenotypic optimum. Unlike the static case, the optimum is no longer constant over time but shifts in a specific direction at a constant average speed.

3.6.1 Geometry of the Movement

Following the theoretical framework established by [3], we assume without loss of generality that the optimum moves along the direction of the first phenotypic component. This simplification is possible because, for isotropic fitness landscapes, any directional movement can be aligned with a principal axis through a change of variables. Consequently, the velocity vector $\mathbf{v}$ is defined with a non-zero magnitude only in its first component (v_1), while all other components remain zero.

3.6.2 Stepwise Dynamics

The movement of the optimum is modeled as a stepwise process rather than continuous motion. The position of the optimum, $z_{opt}(t)$, remains stationary for a fixed period defined by the interval parameter (set to 5,000 simulation steps). At the end of each interval, the optimum shifts to a new position. The magnitude of this jump is determined by the velocity and the interval length:

$$\Delta z = \mathbf{v} \cdot \text{interval} \tag{3.3}$$

3.6.3 Simulation Parameters

To analyze the effect of environmental speed on population fitness, we tested four distinct velocities for the first component:

$$v_1 \in \{0.0032,\ 10^{-4},\ 10^{-5},\ 10^{-8}\} \tag{3.4}$$

The simulations were initialized with a population perfectly adapted to the starting environment (initial mean fitness ≈ 1). We adopted the following configuration to ensure clear visualization of the adaptive dynamics:

- **Duration (T):** The total simulation time was set to $T = 2 \times 10^6$ steps to allow sufficient time for the population to reach a steady state or lag regime.
- **Mutation Rate (μ_{WT}):** We selected a high mutation rate of $\mu_{WT} = 2 \times 10^{-1}$. This value was chosen to enhance the visibility of the population's response.

It is important to note that the mutation rate chosen for these simulations is orders of magnitude higher than typical biological rates. Therefore, the following results presented should be interpreted qualitatively rather than quantitatively, since the high mutation rate ($\mu_{WT} = 0.2$) artificially enhances the population's evolvability, allowing it to track the moving optimum more effectively than a biological population could. Under realistic mutation regimes, the "phenotypic lag" observed in the intermediate velocities would be significantly more pronounced, and the threshold for extinction (Regime 1) would be reached at much lower velocities.

3.6.4 Results

The simulation results reveal four distinct adaptive regimes determined by the velocity of the environmental change (v_1). The population's ability to track the moving optimum depends critically on the balance between the speed of environmental shifts and the rate of genomic adaptation.

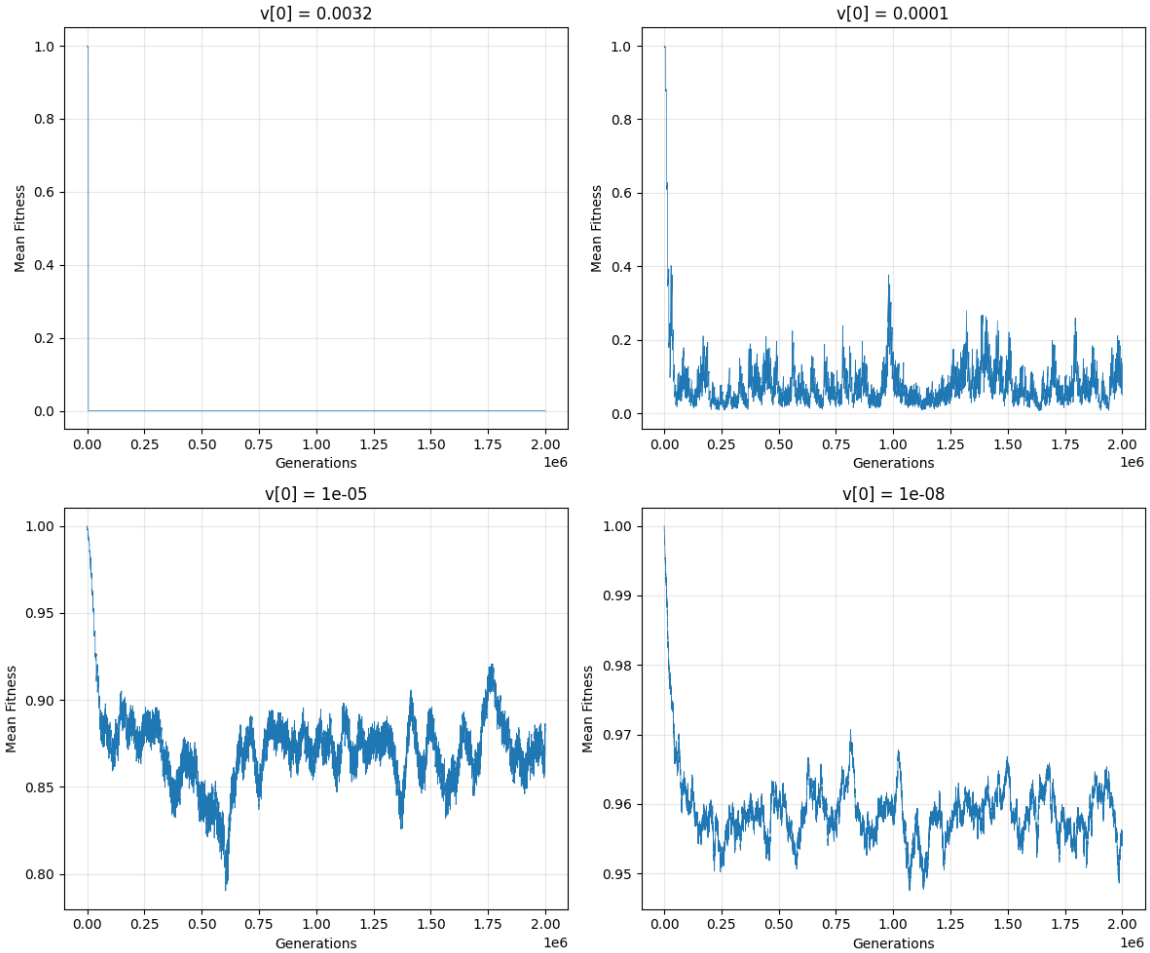


Figure 3.9: Plots of the mean fitness with different values of v

Regime 1: Extinction (High Velocity) At the highest velocities tested ($v_1 = 0.0032$ and $v_1 = 10^{-4}$), the environmental change outpaces the population's adaptive capacity. In the first panel ($v_1 = 0.0032$), the mean fitness collapses rapidly to zero immediately after the initial shifts. The population is unable to generate and fix beneficial mutations fast enough to compensate for the drastic displacement of the optimum. A similar decay is observed in the second panel ($v_1 = 10^{-4}$), confirming that when v exceeds a critical threshold, the population cannot sustain a viable distance from the optimum and adaptation fails.

Regime 2: Near-Perfect Tracking (Low Velocity) At the lowest velocity ($v_1 = 10^{-8}$), the environmental change is slow enough that the population can track the optimum effectively. The fitness remains high, stabilizing in the $[0.85, 0.95]$ range. The "staircase" effect is negligible here, as the adaptive process can almost fully compensate for the small displacements of the optimum.

Regime 3: Dynamic Equilibrium (Intermediate Velocity) The third scenario ($v_1 = 10^{-5}$) is the most interesting: here we observe a different behavior.

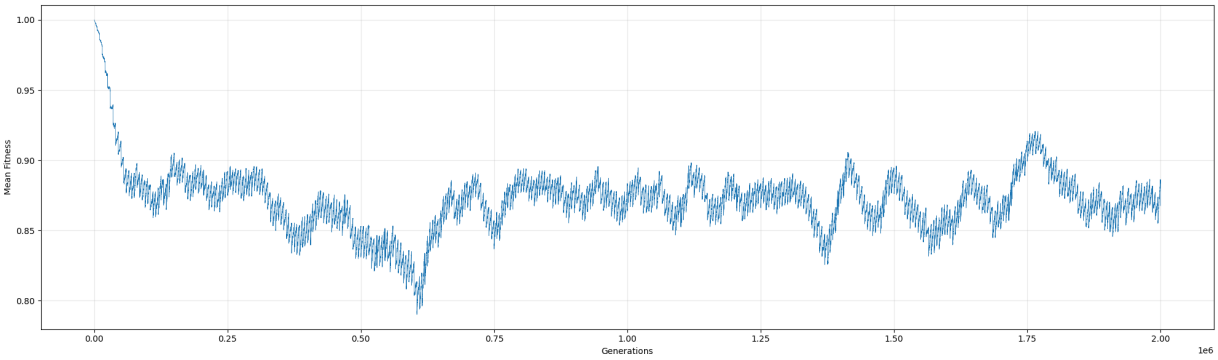


Figure 3.10: Extended plot for $v(0) = 10^{-5}$ for a better visualization.

1. **Initial Decay:** Initially, the mean fitness decreases. This occurs because the population starts perfectly adapted (fitness ≈ 1). In this state, the vast majority of mutations are deleterious, making beneficial mutations rare. As the optimum moves away step-by-step, the fitness drops in a stepwise "staircase" pattern, reflecting the discrete jumps of the optimum.

2. **Stabilization:** As fitness drops, the population effectively moves further from the peak. According to Fisher's Geometric Model, this increases the supply of available beneficial mutations. Eventually, a steady state is reached where the fitness gain from new beneficial mutations exactly balances the fitness loss caused by the moving optimum.

As seen in the plot, the fitness stabilizes within the interval $[0.8, 0.9]$. Importantly, the fitness **does not return to** 1, but instead, it fluctuates around this lower equilibrium value. The population is permanently "lagging" behind the optimum. This behavior observed in the intermediate regimes can be interpreted as a "Red Queen" dynamic. As articulated by Van Valen (1973) and famously referenced in evolutionary theory: *"It takes all the running you can do, to keep in the same place."*

Here, the population is in a constant state of adaptation ("running") to maintain a stable level of fitness ("keeping in the same place"). It never catches up to the optimum; rather, it establishes a phenotypic lag. The magnitude of this lag represents the distance between the population mean and the moving optimum. This results in a permanent reduction in fitness, compared to the optimum, that persists as long as the environment continues to change.

4 Conclusion

In this project, we studied how populations evolve using Fisher's Geometrical Model and a Moran process. We focused on the competition between mutators and wild-type strains, the effect of high mutation rates, and how populations adapt in static and changing environments.

First, our results on mutator competition confirm that the success of a mutator allele depends strongly on its initial frequency. As observed in previous experiments, mutators need to reach a critical frequency to avoid being lost by chance. We also found that mutators are most successful when the population is far from the fitness optimum. When the population is already well-adapted, the advantage of the mutator disappears because beneficial mutations become very rare.

Second, we investigated the effect of the mutation strength, α. We showed that while a higher mutation rate speeds up the early stages of adaptation, it creates a problem in the long term. If the mutation rate is too high, the population cannot stay concentrated at the fitness peak. Instead, it spreads out in the phenotypic space due to a high number of deleterious mutations. This creates a trade-off: a high mutation rate helps to find the optimum, but it makes it harder to stay there efficiently.

Regarding the speed of evolution, our simulations reproduced the "Punctuated Equilibrium" pattern. We observed that evolution proceeds in sudden bursts followed by long periods of stasis. This happens naturally in our model because the population must wait for rare beneficial mutations to appear. We also confirmed the "cost of complexity": as the number of phenotypic dimensions (d) increases, it becomes much harder for the population to adapt, and the rate of fitness increase slows down.

Finally, in the dynamic environment simulations, we identified different regimes based on the speed of the moving optimum. If the environment changes too fast, the population cannot adapt quickly enough and goes extinct. However, at intermediate speeds, we observed a "Red Queen" dynamic. In this scenario, the population constantly improves but never catches up to the optimum. It enters a state of dynamic equilibrium where it is always lagging behind the changing environment.

In conclusion, our findings demonstrate that adaptive evolution is limited by the supply of mutations, the complexity of the organism, and the speed of environmental change. These results help to clarify the conditions under which mutator alleles can fix and how populations survive in unstable environments.

References

[1] Lin Chao and Edward C. Cox. Competition between high and low mutating strains of *Escherichia coli*. *Evolution*, 37(1):125–134, 1983.

[2] Olivier Tenaillon, O. K. Silander, J.-P. Uzan, and Lin Chao. Quantifying organismal complexity using a population genetic approach. *PLoS ONE*, 2(2):e217, 2007.

[3] Sebastian Matuszewski, Joachim Hermisson, and Michael Kopp. Fisher's geometric model with a moving optimum. *Evolution*, 68(9):2571–2588, 2014.

[4] K. Sprouffske, J. Aguilar-Rodríguez, P. Sniegowski, and A. Wagner. High mutation rates limit evolutionary adaptation in *Escherichia coli*. *PLoS Genetics*, 14(4):e1007324, 2018.

[5] Indranath Chaudhuri and Indrani Bose. Punctuated equilibrium in an evolving bacterial population. *Physica A: Statistical Mechanics and its Applications*, 270(1–2):63–68, August 1999.

[6] João A. Moura de Sousa, João Alpedrinha, Paulo R. A. Campos, and Isabel Gordo. Competition and fixation of cohorts of adaptive mutations under fisher geometrical model. *PeerJ*, 4:e2256, 2016.